

31	27	26	25	24	20	19	15	14	12	11	7	6	0	
funct7				rs2		rs1		funct3		rd		opcode		R-type
imm[11:0]						rs1		funct3		rd		opcode		I-type
imm[11:5]				rs2		rs1		funct3		imm[4:0]		opcode		S-type
imm[12 10:5]				rs2		rs1		funct3		imm[4:1 11]		opcode		B-type
imm[31:12]										rd		opcode		U-type
imm[20 10:1 11 19:12]										rd		opcode		J-type

RV32I Base Instruction Set									
imm[31:12]					rd	0110111	LUI		
imm[31:12]					rd	0010111	AUIPC		
imm[20 10:1 11 19:12]					rd	1101111	JAL		
imm[11:0]			rs1	000	rd	1100111	JALR		
imm[12 10:5]		rs2	rs1	000	imm[4:1 11]	1100011	BEQ		
imm[12 10:5]		rs2	rs1	001	imm[4:1 11]	1100011	BNE		
imm[12 10:5]		rs2	rs1	100	imm[4:1 11]	1100011	BLT		
imm[12 10:5]		rs2	rs1	101	imm[4:1 11]	1100011	BGE		
imm[12 10:5]		rs2	rs1	110	imm[4:1 11]	1100011	BLTU		
imm[12 10:5]		rs2	rs1	111	imm[4:1 11]	1100011	BGEU		
imm[11:0]			rs1	000	rd	0000011	LB		
imm[11:0]			rs1	001	rd	0000011	LH		
imm[11:0]			rs1	010	rd	0000011	LW		
imm[11:0]			rs1	100	rd	0000011	LBU		
imm[11:0]			rs1	101	rd	0000011	LHU		
imm[11:5]		rs2	rs1	000	imm[4:0]	0100011	SB		
imm[11:5]		rs2	rs1	001	imm[4:0]	0100011	SH		
imm[11:5]		rs2	rs1	010	imm[4:0]	0100011	SW		
imm[11:0]			rs1	000	rd	0010011	ADDI		
imm[11:0]			rs1	010	rd	0010011	SLTI		
imm[11:0]			rs1	011	rd	0010011	SLTIU		
imm[11:0]			rs1	100	rd	0010011	XORI		
imm[11:0]			rs1	110	rd	0010011	ORI		
imm[11:0]			rs1	111	rd	0010011	ANDI		
0000000		shamt	rs1	001	rd	0010011	SLLI		
0000000		shamt	rs1	101	rd	0010011	SRLI		
0100000		shamt	rs1	101	rd	0010011	SRAI		
0000000		rs2	rs1	000	rd	0110011	ADD		
0100000		rs2	rs1	000	rd	0110011	SUB		
0000000		rs2	rs1	001	rd	0110011	SLL		
0000000		rs2	rs1	010	rd	0110011	SLT		
0000000		rs2	rs1	011	rd	0110011	SLTU		
0000000		rs2	rs1	100	rd	0110011	XOR		
0000000		rs2	rs1	101	rd	0110011	SRL		
0100000		rs2	rs1	101	rd	0110011	SRA		
0000000		rs2	rs1	110	rd	0110011	OR		
0000000		rs2	rs1	111	rd	0110011	AND		
fm	pred	succ	rs1	000	rd	0001111	FENCE		
1000	0011	0011	00000	000	00000	0001111	FENCE.TSO		
0000	0001	0000	00000	000	00000	0001111	PAUSE		
000000000000			00000	000	00000	1110011	ECALL		
000000000001			00000	000	00000	1110011	EBREAK		